Algorithm Design - Second Midterm

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Instructions:

- Before you start, write on the top of each sheet (1) your full name, (2) your student ID (*matricola*) and (3) the sheet number (1, 2, 3, and 4).
- If you use additional sheets as rough draft (*brutta*) for calculations, etc., you must hand it, along with the rest of the exam. You can keep the question sheet.
- The rooms are small so it may be tempting to copy. Note that if we catch you copying in any way (copying from another student, using your phone, etc.) you will be automatically disqualified.

Problem 1

(10 points) Consider the following decision problem.

Clique: Given an undirected graph $G = (V, E)$ and an integer k, does there exist a subset of vertices $S \subseteq V$ with $|S| \geq k$ that induces a complete subgraph, i.e., a *clique*?

1. **(2 points)** Define the notion of polynomial time reduction and NP completeness;
2. **(2 points)** Prove that Clique is in NP;
3. **(6 points)** Prove that Clique is NP-complete. A direct reduction from 3-SAT is preferable, alternatively you can prove a 2-step reduction from 3-SAT via Independent Set.

Problem 2

(10 points) Consider the following Load Balancing problem. There are m identical machines and n jobs where job j has processing time t_j. One job has to run on one machine without interruption and a machine can process at most one job at a time. If $J(i)$ is the subsets of jobs assigned to machine i, the *load* is defined as

$$L_i = \sum_{j \in J(i)} t_j.$$

The Load Balancing problems asks to assign each job to a machine to minimize the makespan i.e. the $\max_i L_i$.

1. **(3 points)** Consider the decision version of Load Balancing, that is, given an integer T, decide whether there is a schedule with makespan $\leq T$. Prove that this problem is NP-complete.
2. **(3 points)** Prove that the List Scheduling algorithm (assigning job j to the machine whose load is smallest so far) achieves a 2 approximation ratio;
3. **(4 points)** Prove that the Longest Processing Time (LPT) scheduling that (1) sorts the jobs in descending order of processing time, and then (2) uses List Scheduling, achieves a 3/2 approximation ratio.

Problem 3

(10 + 5 points) Consider the Directed Max-Cut problem: given a directed graph $G = (V, A)$ with non-negative arc weights w_{ij}, find a cut $(S, V \setminus S)$ that maximizes

$$\sum_{\substack{(i,j) \in A \\ i \in S,\ j \notin S}} w_{ij}.$$

1. **(5 points)** Show a randomized algorithm that achieves an expected approximation ratio of $\frac{1}{4}$;
2. **(3 points)** Show an equivalent deterministic algorithm obtained by derandomizing the proposed procedure;

3. **(2 points)** Prove that the following integer program models the problem:

$$\begin{aligned}
\max \sum_{(i,j)\in A} & w_{ij} z_{ij} & \\
\text{s.t. } & z_{ij} \le x_i, & \forall (i,j) \in A, \\
& z_{ij} \le 1 - x_j, & \forall (i,j) \in A, \\
& x_i \in \{0,1\}, & \forall i \in V, \\
& 0 \le z_{ij} \le 1, & \forall (i,j) \in A.
\end{aligned}$$

4. **(Bonus, 5 points)** Consider the randomized rounding algorithm that finds the optimal solution to the LP relaxation of the above program, and then assigns i to S with probability $\left(\frac{1}{4} + \frac{x_i}{2}\right)$. Show that it achieves an expected approximation ratio of $\frac{1}{2}$.